

A surface charge density: $\rho_s = ne\Delta x$

The electric field $E_x = \frac{ne\Delta x}{\epsilon\epsilon_0}$

The electric The electric field acts back on the electrons

$$-eE_x = -\frac{ne^2\Delta x}{\epsilon\epsilon_0} = m_e \frac{\partial^2}{\partial t^2} (\Delta x)$$

Which is a harmonic oscillation with frequency:

$$\omega_{PL}^0 = \left(\frac{ne^2}{m_e \epsilon\epsilon_0} \right)^{\frac{1}{2}} = \omega_L, \omega_T = 0 \quad k \rightarrow 0$$

If we go to shorter wavelengths we find a weak parabolic dispersion for the plasmons

$$\omega_{PL}(k) = \omega_{PL}^0(1 + ak^2 + \dots)$$

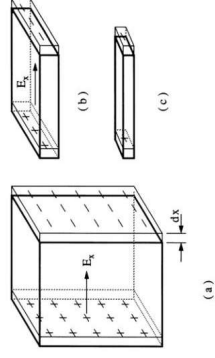


Fig. 10.1. Schematic drawing of a displacement of the electron system in a plasma oscillation in a three- (a), two- (b) and one-dimensional semiconductor (c)

CHAPTER 10

Plasmons and Magnons

Contents

- In this chapter we will briefly address some other collective excitations in semiconductors and the quasi-particles which result from the quantization of these excitations like plasmons or magnons
- Considering a semiconductor which contains a large number of electrons ($10^{17} - 10^{19} \text{cm}^{-3}$), collective oscillations of electrons $\rightarrow$ Plasma oscillations
- Magnons, existing in ferro-, ferri- and antiferromagnetic materials, the collective excitations of the spin system

Plasma oscillation-electron collective oscillation

- Only longitudinal oscillations can propagate
- The quanta of plasma oscillation is plasmon
- The plasmons obey Bose statistics
- For large wavevectors: the plasmon modes are strongly damped
- For semiconductors, the plasmon energy is 10-100 meV
- For metals, the plasma frequency is usually in Vis-UV range



$$\begin{aligned}\varepsilon(\omega) &= 1 + \frac{Ne^2}{\varepsilon_0 m} (\omega_p^2 - \omega^2 - i\omega\gamma)^{-1} \\ \varepsilon(\omega) &= \varepsilon_b + \frac{\omega_p^2}{\omega^2 - \omega^2 - i\omega\gamma}\end{aligned}$$



$$\varepsilon(\omega) = \varepsilon_b + \frac{ne^2}{m_e \varepsilon_0} \frac{1}{-\omega^2 - i\omega\gamma} \quad (10.3)$$

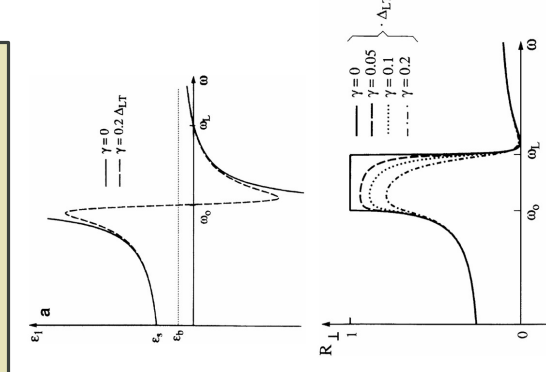


Fig. 10.4 The dielectric function of a plasmon resonance (a) and the resulting reflection spectrum (b) for vanishing damping

The equation of motion for free carriers for an electric field :

$$m\ddot{x} = -\frac{m}{\tau}\dot{x} + eE_x^0 e^{-i\omega t}$$

$$(10.4a)$$

Resulting in a frequency-dependent electrical conductivity:

$$\sigma(\omega) = \sigma_0 \frac{1}{1 - i\omega\tau}$$

$$(10.4b)$$

and to a total dielectric function:

$$\varepsilon_{total}(\omega) = \varepsilon_{Lorentz}(\omega) + \chi_{free\ carrier}(\omega)$$

$$(10.4c)$$

with the help of Maxwells equation $\nabla \times \mathbf{H} = \mathbf{j} + (\varepsilon_0 \dot{\mathbf{E}} + \dot{\mathbf{P}})$ $\nabla \times \mathbf{H} = \mathbf{j} + \dot{\mathbf{D}} = \sigma \mathbf{E} + \varepsilon_0 \varepsilon_{Lorentz} \dot{\mathbf{E}}$
we obtain :

$$\chi_{free\ carrier}(\omega) = \frac{i}{\varepsilon_0 \omega} \sigma(\omega)$$

and for the total dielectric function:

$$\varepsilon_{total}(\omega) = \varepsilon_{Lorentz}(\omega) + \frac{i}{\varepsilon_0 \omega} \sigma(\omega) = \varepsilon_{Lorentz}(\omega) + \frac{\omega_{pl}^2}{-\omega^2 - i\omega/\tau}$$

$$\varepsilon(\omega) = \varepsilon_b + \frac{ne^2}{m\varepsilon_0} \frac{1}{-\omega^2 - i\omega\gamma}$$

Quantized (surface) plasmon modes exist in small metal spheres or colloids.